

MESTTS SESSION REPORT

Multimodal Eye–Speech–Typing Temporal Synchronization System

Participant ID	p19	Session ID	3ebe9133-cdd2-47
Date / Time	2026-04-12 09:57	Sentences Scored	3

DYSLEXIA RISK ASSESSMENT

MODERATE

Risk Score: 0.3500

EVS Gaze Span	Calibration Quality	Data Completeness	Sentences Scored
0.0884	45.0%	81.9%	3

Extracted Feature Vector

F01_evs_speech_std_ms	F02_mean_evs_speech_ms	F03_evs_speech_skew	F04_speech_inversion_.
442.5000	197.1667	0.0000	0.5000
F05_evs_typing_std_ms	F06_mean_fixation_dur.	F07_std_fixation_dura.	F08_regression_rate
1438.1911	180.9442	14.1127	0.1111
F09_regression_count_.	F10_fixation_outlier_.	F11_typing_inversion_.	F12_mean_correction_r.
2	1.1028	0.0556	0.0000
F13_max_corrections	F14_stg_std_ms	F15_stg_negative_rate	F17_corr_evs_typing_l.
0	137.8333	0.0000	-0.2989

IMPORTANT: This system produces a behavioral risk indicator only. It does not constitute a clinical diagnosis of dyslexia or any other condition. Results must be interpreted by a qualified specialist in the context of a comprehensive assessment protocol.

Interpretation

Overall Assessment

Some markers of reading difficulty were detected — further assessment recommended.

The participant's score of 0.350 places them in the moderate-risk band. Certain reading behaviours — such as irregular fixation timing, gaze-speech misalignment, or variable reading pace — appeared more often than typical. A follow-up clinical assessment would be worthwhile.

Data Quality

EVS Gaze Span — LOW

Span of 0.0884 is below the recommended threshold (0.10). Consider re-running calibration with more exaggerated eye movement.

Calibration Quality — MODERATE

45% quality suggests the calibration was acceptable but not ideal. Word-level mapping may have minor errors.

Data Completeness — HIGH

82% of word slots had both gaze and speech data captured. The session data is highly reliable.

Key Reading Behaviour Indicators

Eye-Voice Span (F02)

The eye typically looked ahead of spoken words by ~197 ms. This is normal — fluent readers pre-scan upcoming words.

EVS Consistency (F01)

EVS variability was moderate (442 ms std). Some inconsistency in reading pace.

Mean Fixation Duration (F06)

Mean fixation of 181 ms is in the typical range (150-280 ms). The participant paused a normal amount of time on each word.

Regression Rate (F08)

Regression rate of 11.1% is mild. Some re-reading occurred.

Gaze-Speech Correlation (F16)

Gaze-speech correlation of 0.000 is very low or negative. The participant's eye movements and speech were poorly synchronised.

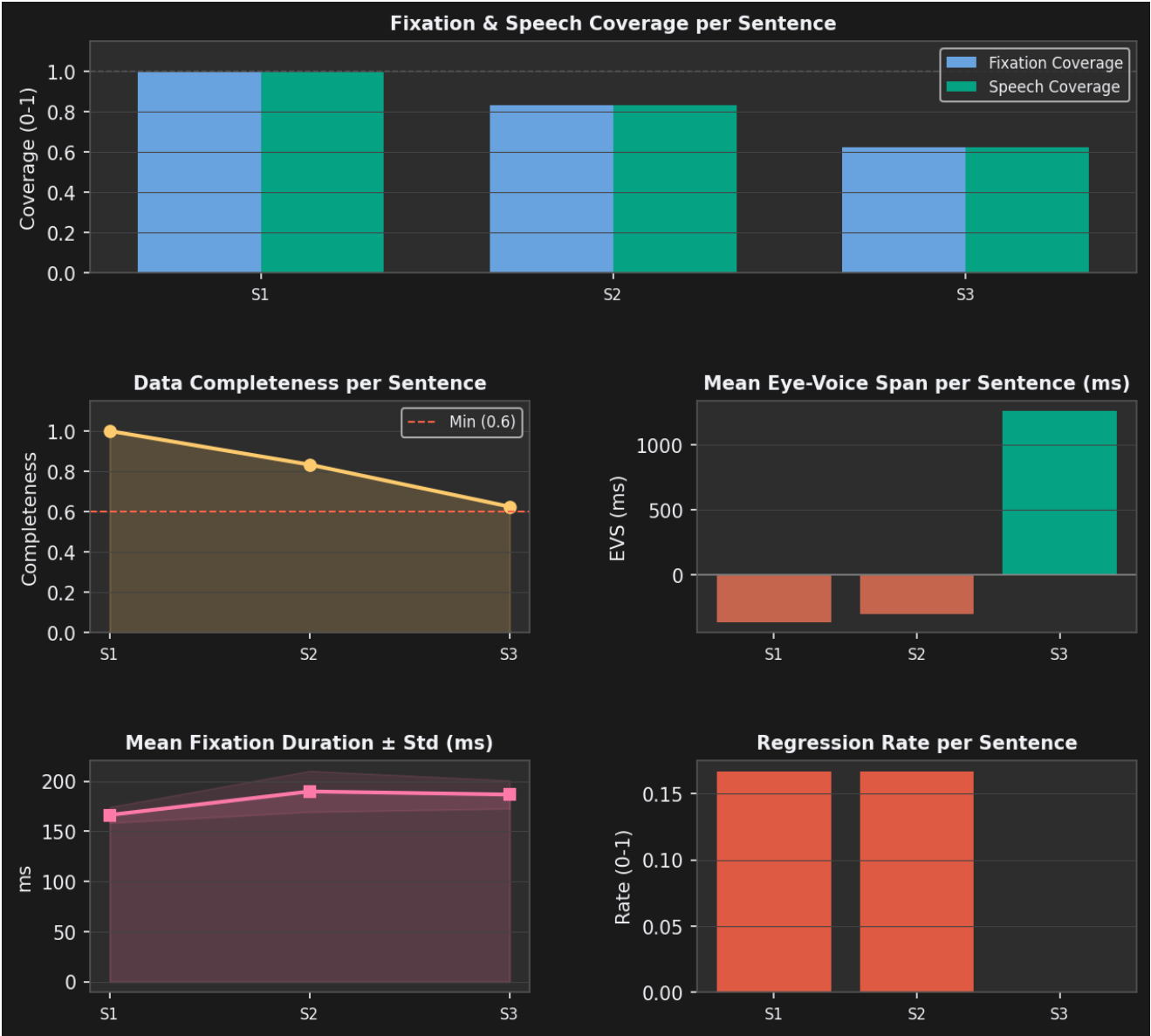
What to Do Next

Next Steps

Share these results with a qualified educational psychologist or specialist reading teacher. A formal psychoeducational or phonological assessment is recommended to clarify whether intervention is needed. Interim classroom supports may be worth exploring while awaiting formal assessment.

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Per-Sentence Analysis



Typing Analysis



Words Typed	Typing Coverage	Total Corrections	Mean STG	Mean EVS Typing
20 / 20	100%	0	3868 ms	4671 ms

Feature Vector

Feature	Value	Description
F01_evs_speech_std_ms	442.5000	EVS Speech variability (std, ms)
F02_mean_evs_speech_ms	197.1667	Mean EVS Speech (ms)
F03_evs_speech_skew	0.0000	EVS Speech skewness
F04_speech_inversion_rate	0.5000	Speech inversion rate (voice leads eye)
F05_evs_typing_std_ms	1438.1911	EVS Typing variability (std, ms)
F06_mean_fixation_duration_ms	180.9442	Mean fixation duration (ms)
F07_std_fixation_duration_ms	14.1127	Fixation duration std (ms)
F08_regression_rate	0.1111	Regression rate (re-reads / word)
F09_regression_count_total	2	Total regression count
F10_fixation_outlier_ratio	1.1028	Fixation outlier ratio (max/mean)
F11_typing_inversion_rate	0.0556	Typing inversion rate
F12_mean_correction_rate	0.0000	Mean correction rate (backspaces/word)
F13_max_corrections	0	Max corrections in any single word
F14_stg_std_ms	137.8333	Speech-Typing Gap std (ms)
F15_stg_negative_rate	0.0000	STG negative rate
F17_corr_evs_typing_latency	-0.2989	EVS-typing latency correlation

